

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 19th February, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 2	CLOs to be covered: CLO1
Lab Title: For Loops	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 2: For Loops

Lab Goals/Objectives:

To understand and apply the for loop construct in Python to iterate over sequences and repeat tasks a fixed number of times.

Equipment Required:

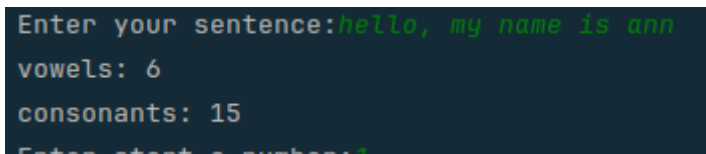
Python 3, VS Code / PyCharm

Theory:

The for loop in Python is used to iterate over a sequence (such as a string, list, or range of numbers) and execute a block of code once for each item in that sequence.

Lab Work:**Task 1: Vowels and Consonants**

```
sentence= input("Enter your sentence:")
vowels =0
consonants =0
i=0
for i in range (len(sentence)):
    if sentence[i] in 'aeiou'or sentence[i] in 'AEIOU':
        vowels +=1
    else:
        consonants +=1
print("vowels:",vowels)
print("consonants:",consonants)
```

Result:

```
Enter your sentence:hello, my name is ann
vowels: 6
consonants: 15
Enter start a number:
```

Task 2: Average GPA

```
subjects=int(input("Enter number of subjects: "))
print("Total Subject are",subjects)
total_credit_hours =0
total_grade_points =0
for i in range(1,subjects+1):
    gp = int(input("Enter Grade Points: "))
    credit = int(input("Enter credit hours: "))

    total_credit_hours += credit
    total_grade_points += credit*gp
print("Your credit hours are",total_credit_hours)
print("Your grade points are",total_grade_points)
if total_grade_points >0:
    GPA=(total_grade_points/total_credit_hours)
    print("Your GPA for this semester is ",GPA)
else:
    print("You have not earned any credits for this semester")
```

Result:

```
Total Subject are 2
Enter Grade Points: 4
Enter credit hours: 2
Enter Grade Points: 4
Enter credit hours: 3
Your credit hours are 5
Your grade points are 20
Your GPA for this semester is 4.0
```

Task 3: Display prime numbers in the range and their sum

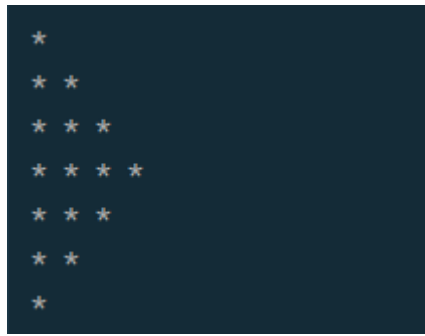
```
start= int(input("Enter start a number:"))
end= int(input("Enter end a number:"))
sum=0
for i in range(start,end+1):
    for j in range(2,i):
        if i%j==0:
            break
    else:
        print(i)
        sum +=i
print("sum:",sum)
```

Result:

```
Enter start a number:1
Enter end a number:10
1
2
3
5
7
sum: 10
```

#Task 4: Draw a pattern

```
n=4
for i in range(n):
    for j in range(i+1):
        print("*",end=" ")
    print()
r=3
for i in range(r):
    for j in range(i,r):
        print("*",end=" ")
    print()
```

Result:**Conclusions:**

The for loop was used to solve counting and accumulation problems, reinforcing the concept of iteration.